



Class XI - Computer Science – Practice Worksheet

◆ Python Identifiers & Keywords 3 Q

Q1. Which of the following is a valid identifier in Python?

- A. _myVar
- B. 1stValue
- C. my-var
- D. import

Q2. Which of the following is **not** a Python keyword?

- A. continue
- B. break
- C. define
- D. return

Q3. Which of the following is a valid identifier in Python?

- A. student_name
- B. student-name
- C. 2student
- D. while

◆ Input & Output Functions 3 Q

Q4. What will `print('Welcome', 'Students', sep='-', end='!')` display?

- A. Welcome-Students!
- B. Welcome Students!
- C. Welcome-Students
- D. Welcome Students

Q5. What does `input('Enter your name:')` always return when user types "John"?

- A. int
- B. float
- C. string
- D. boolean

Q6. What will `print('Hello', end='***')` display?

- A. Hello***
- B. Hello
- C. ***Hello
- D. Error

◆ Built-in Functions 9 Q

Q7. What will `len('Computer Science')` return?

- A. 15
- B. 16
- C. 17
- D. 18

Q8. What is the output of `list(range(5, 15, 3))`?

- A. [5, 8, 11, 14]
- B. [5, 8, 11, 14, 17]
- C. [5, 6, 7, 8, 9, 10, 11, 12, 13, 14]
- D. [8, 11, 14]

Q9. What will be the output of `min([15, 8, 23, 5, 12]) + max([15, 8, 23, 5, 12])`?

- A. 23
- B. 28
- C. 35
- D. 38

Q10. What does `len([10, 20, 30, 40, 50, 60])` return?

- A. 4
- B. 5
- C. 6
- D. 7

Q11. What will `sum([2, 4, 6, 8, 10])` return?

- A. 20
- B. 25
- C. 30
- D. 35

Q12. What is the output of this code?

```
x = '456'  
print(type(int(x)))
```

- A. `<class 'str'>`
- B. `<class 'int'>`
- C. `<class 'float'>`
- D. Error

Q13. What will `range(2, 15, 4)` generate?

- A. [2, 6, 10, 14]
- B. [2, 6, 10, 14, 18]
- C. [2, 4, 6, 8, 10, 12, 14]

D. [4, 8, 12]

Q14. What is the default stop value in range(8)?

- A. 0
- B. 7
- C. 8
- D. 9

Q15. What will type(3.14) return?

- A. <class 'int'>
- B. <class 'float'>
- C. <class 'str'>
- D. <class 'bool'>

◆ Conditional Statements (if, elif, else) 9 Q

Q16. Which statement is correct to check if y is less than 20?

- A. if y < 20:
- B. if (y =< 20):
- C. if y lesser 20:
- D. if y << 20:

Q17. What does this code print?

```
if 'e' in 'elephant':  
    print('Found')  
else:  
    print('Not Found')
```

- A. Found
- B. Not Found
- C. Error
- D. Nothing

Q18. What will happen if both **if** and **elif** conditions are false?

- A. Both blocks execute
- B. Only if executes
- C. Only elif executes
- D. else block executes

Q19. Consider this code:

```
n = 20
if n % 2 == 0:
    if n % 5 == 0:
        print('Divisible by 2 and 5')
    else:
        print('Divisible by 2 only')
else:
    print('Not divisible by 2')
```

- A. Divisible by 2 only
- B. Divisible by 2 and 5
- C. Not divisible by 2
- D. Error

Q20. Which of the following correctly checks if x is not equal to y?

- A. if x != y:
- B. if x =! y:
- C. if x not = y:
- D. if (x : y):

Q21. What's the output of this snippet?

```
num = 8
if num % 2 == 0:
    print('Even')
else:
    print('Odd')
```

- A. Even
- B. Odd
- C. 8
- D. Error

Q22. What will this code print?

```
score = 75
if score >= 90:
    print("A")
elif score >= 75:
    print("B")
else:
    print("C")
```

- A. A
- B. B
- C. C
- D. Error

Q23. What is the output?

```
x = 15
if x > 10:
    print("Big")
if x > 20:
    print("Huge")
else:
    print("Small")
```

- A. Big
- B. Big Small
- C. Big Huge
- D. Big Huge Small

Q24. What will this code print?

```
a = 25
if a > 20:
    print("A")
elif a > 15:
    print("B")
else:
    print("C")
```

- A. A
- B. B
- C. C
- D. Error

◆ Looping Statements (for, while) 8 Q

Q25. What is printed by this code?

```
for i in range(4, 9):  
    print(i, end=' ')
```

- A. 4 5 6 7 8
- B. 4 5 6 7 8 9
- C. 0 1 2 3 4 5 6 7 8
- D. Error

Q26. Predict the output:

```
for i in range(2, 12, 4):  
    print(i, end=' ')
```

- A. 2 6 10
- B. 2 4 6 8 10
- C. 2 6 10 14
- D. 2 4 6 8 10 12

Q27. What will this code print?

```
for i in range(5, 0, -2):  
    print(i, end=' ')
```

- A. 5 3 1
- B. 5 4 3 2 1
- C. 5 3 1 -1
- D. Error

Q28. What is the output?

```
count = 1  
while count <= 4:  
    print(count, end=' ')  
    count += 2
```

- A. 1 3

- B. 1 3 5
- C. 1 2 3 4
- D. Error

Q29. How many times will the loop execute?

```
for i in range(3, 13, 5):  
    print(i)
```

- A. 2
- B. 3
- C. 4
- D. 5

Q30. What will this code print?

```
x = 3  
while x < 10:  
    print(x, end=' ')  
    x = x + 2
```

- A. 3 5 7 9
- B. 3 5 7 9 11
- C. 3 4 5 6 7 8 9
- D. 3 5 7

Q31. What is the output?

```
for i in range(10, 0, -3):  
    print(i, end=' ')
```

- A. 10 7 4 1
- B. 10 7 4 1 -2
- C. 10 9 8 7 6 5 4 3 2 1
- D. 7 4 1

Q32. How many times will the while loop execute?

```
n = 5  
while n < 15:
```

```
print(n)  
n = n + 2
```

- A. 4
- B. 5
- C. 6
- D. 7

◆ Mutable & Immutable Data Types 5 Q

Q33. Which of the following is a mutable data type in Python?

- A. int
- B. str
- C. list
- D. tuple

Q34. Which of the following is an immutable data type?

- A. list
- B. dict
- C. set
- D. tuple

Q35. Which of the following data types is mutable?

- A. str
- B. int
- C. float
- D. list

Q36. Which of the following is immutable?

- A. list
- B. dict
- C. tuple

D. set

Q37. Which of the following data types is mutable?

- A. tuple
- B. str
- C. dict
- D. float

◆ Types of Statements 3 Q

Q38. Which of the following is a compound statement in Python?

- A. `x = 10`
- B. `if x > 5:`
 `print("Yes")`
- C. `print("Hello")`
- D. `import math`

Q39. What type of statement is `x = 5 + 3`?

- A. Simple statement
- B. Compound statement
- C. Conditional statement
- D. Iterative statement

Q40. What type of statement is the following?

```
for i in range(5):  
    print(i)
```

- A. Simple statement
- B. Compound statement
- C. Expression statement
- D. Assignment statement

◆ Membership Operators (in, not in) 4 Q

Q41. What will this code print?

```
colors = ['red', 'blue', 'green', 'yellow']
if 'purple' in colors:
    print("Present")
else:
    print("Absent")
```

- A. Present
- B. Absent
- C. Error
- D. Nothing

Q42. What is the output?

```
word = "education"
if 'cat' in word:
    print("Yes")
else:
    print("No")
```

- A. Yes
- B. No
- C. Error
- D. Nothing

Q43. What will this code print?

```
numbers = [1, 2, 3, 4, 5]
if 6 not in numbers:
    print("Not found")
else:
    print("Found")
```

- A. Not found
- B. Found
- C. Error
- D. Nothing

Q44. What will this code print?

```
text = "programming"
if 'pro' in text:
    print("Contains")
else:
    print("Not contains")
```

- A. Contains
- B. Not contains
- C. Error
- D. Nothing

◆ Output Prediction – Conditional + Loop 13 Q

Q45. Predict the output:

```
x = 12
y = 8
if x % y == 4:
    print(x)
else:
    print(y)
```

Q46. Predict the output:

```
values = [3, 6, 9, 12, 15]
total = 0
for val in values:
    if val % 3 == 0 and val < 12:
        total += val
print(total)
```

Q47. Predict the output:

```
data = [5, 15, 25, 35, 45]
for index in range(len(data)):
    if index == 3:
        print(data[index] * 2)
```

Q48. Predict the output:

```
color = 'blue'
sky = color
if sky == 'blue':
    a = 5
    print(color)
else:
    a = 3
    b = a + 2
    print(sky, color)
```

Q49. Predict the output:

```
if '2024' in 'Calendar':
    print("YES")
else:
    print("NO")
```

Q50. Predict the output:

```
total = 0
for num in range(-3, 4):
    if num:
        total += num
print(total)
```

Q51. Predict the output:

```
numbers = [2, 4, 6, 8, 10, 12]
result = 0
for num in numbers:
    if num % 4 == 0 and num > 5:
        result += num
print(result)
```

Q52. Predict the output:

```
for x in range(15, 60, 25):  
    y = x  
    while y % 5 == 0:  
        print(x)  
        y = y + 1
```

Q53. Predict the output:

```
items = [7, 3, 9, 1, 5]  
if 8 in items:  
    print(True)  
else:  
    print(False)  
print("Done")
```

Q54. Predict the output:

```
a = '8923'  
b = 5  
if a == b:  
    print(b)  
else:  
    print(a)
```

Q55. Predict the output:

```
m = 7  
n = 5  
if m > n:  
    print(m * 2)  
else:  
    print(n * 3)
```

Q56. Predict the output:

```
nums = [10, 20, 30, 40, 50]
sum_val = 0
for i in nums:
    if i % 20 == 0:
        sum_val += i
print(sum_val)
```

Q57. Predict the output:

```
p = 8
q = 12
if p * 2 == q:
    print(p + q)
else:
    print(q - p)
```

◆ Selection / Conditional Statements 3 Q

Q58. What is the syntax for a multi-way selection statement in Python?

- A. if
- B. if-else
- C. if-elif-else
- D. while

Q59. What is the purpose of the `elif` statement in Python?

- A. To start a loop
- B. To handle multiple conditions
- C. To define a function

D. To import a module

Q60. How many `else` statements can be used with a single `if` statement?

- A. 0
- B. 1
- C. 2
- D. Unlimited

◆ Iterative / Looping Statements 3 Q

Q61. Which loop in Python is used when the number of iterations is known?

- A. for loop
- B. while loop
- C. if loop
- D. else loop

Q62. What is the correct syntax for a `while` loop in Python?

- A. `while x < 5:`
- B. `while (x < 5)`
- C. `while x < 5 do:`
- D. `while x < 5:`

Q63. Which loop in Python is used when the number of iterations is unknown?

- A. for loop
- B. while loop
- C. if loop
- D. else loop

◆ Error Correction & Debugging 6 Q

Q64. Rewrite the following code after removing all syntax error(s):

```
20 = Limit
For i in range(1, Limit)
    IF i % 3 == 0: print i * 3
    Else: print i + 2
```

(Write corrected code in your answer sheet)

Q65. Identify and correct the errors (syntax & logical) in the following code:

```
Start = 50
while start > 10
    print(start, end = " ")
    start - 2
print("Process Complete")
```

(Write corrected code in your answer sheet)

Q66. Correct the errors in the following code:

```
value = 7
if value = 7:
    printf("Match")
else:
    print("No Match")
```

(Write corrected code in your answer sheet)

Q67. Find and correct the errors:

```
count = 0
while count < 6
    print(count)
    count = count + 1
print("Loop finished")
```

(Write corrected code in your answer sheet)

Q68. Correct the errors:

```
num = 10
if num > 5
    print("Greater")
else
    print("Smaller")
```

(Write corrected code in your answer sheet)

Q69. Correct the errors:

```
x = 15
if x < 20:
    print("Less")
elif x > 10:
    print("Between")
else:
    print("Greater")
```

(Write corrected code in your answer sheet)

◆ Factorial Program 3 Q

Q70. Write a Python code to find the Factorial of a number using a while loop.

Q71. Write a Python code to find the Factorial of a number using a for loop.

Q72. Write a Python code to find the Factorial of a number entered by the user.

◆ Fibonacci Series 3 Q

Q73. Write a Python code to generate the first 15 Fibonacci Series

numbers.

Q74. Write a Python program to generate Fibonacci series up to N terms using a `while` loop.

Q75. Write a Python code to generate the first 10 Fibonacci Series numbers using a `for` loop.

◆ Sum of Natural Numbers & Multiplication Table

4
Q

Q76. Write a Python program to do the following:

- i. Find the sum of first N natural numbers using a `for` loop.
- ii. Print the multiplication table of a number using a `while` loop.

Q77. Write a Python program to find the sum of squares of first N natural numbers.

Q78. Write a Python program to print the multiplication table of all numbers from 1 to 5.

Q79. Write a Python program to find the sum of first N natural numbers using a `while` loop.

◆ Bank Operations Menu

3 Q

Q80. Write a Python program to perform Bank Operations Menu (using any type of IF Statement):

Input: Account_Number, Customer_Name, Current_Balance, Branch_Code

Menu 1. Deposit – If choice is 1: input deposit amount, add to balance, display new balance.

Menu 2. Withdrawal – If choice is 2: input withdrawal amount, deduct from balance, display new balance.

Menu 3. Balance Enquiry – If choice is 3: display Account_Number, Customer_Name, and Current_Balance.

Q81. Write a Python program for a simple calculator with menu options (Add, Subtract, Multiply, Divide) using if-elif-else.

Q82. Write a Python program for a library management menu system with options to issue, return, and display books using if-elif-else.

◆ Product Discount Calculation (if-elif-else) 4 Q

Q83. Write a Python code using IF...ELIF...ELSE:

Input: Item_Code, Item_Name, Quantity, Rate.

Calculate Discount:

- Rate $\geq$ 25000 $\rightarrow$ Discount 40%
- Rate $\geq$ 15000 $\rightarrow$ Discount 25%
- Rate $\geq$ 8000 $\rightarrow$ Discount 12%

Print: Item_Code, Item_Name, Quantity, Rate, Discount, and
Total_Amount = (Quantity $\times$ Rate) – Discount.

Q84. Write a Python program to calculate the grade of a student based on percentage using if-elif-else.

Q85. Write a Python program to calculate the electricity bill based on units consumed using if-elif-else.

Q86. Write a Python program to calculate the tax based on income using if-elif-else.